

Dan Wang

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Web: <https://xjtu-netverify.github.io/people/dwang/>

Research Interests

- Networking: network verification through symbolic and/or distributed network simulation.
- Automated reasoning: access control policy analysis on public clouds.

Publications

- [ASE'25] **Dan Wang**, Peng Zhang, Zhenrong Gu, Weibo Lin, Shibiao Jiang, Zhu He, Xu Du, Longfei Chen, Jun Li, Xiaohong Guan. "RELIA: Accelerating Analysis of Cloud Access Control Policies." Proceedings of the IEEE/ACM ASE 2025.
- [SIGCOMM'25] **Dan Wang**, Peng Zhang, Wenbing Sun, Wenkai Li, Xing Feng, Hao Li, Jiawei Chen, Weirong Jiang, and Yongping Tang. "S2: A Distributed Configuration Verifier for Hyper-Scale Networks." Proceedings of the ACM SIGCOMM 2025.
- [SIGCOMM'24] **Dan Wang**, Peng Zhang, and Aaron Gember-Jacobson. "Expresso: Comprehensively Reasoning About External Routes Using Symbolic Simulation." Proceedings of the ACM SIGCOMM 2024. (**Best Paper Honorable Mention**)
- [SIGCOMM'22] Peng Zhang, **Dan Wang**, and Aaron Gember-Jacobson. "Symbolic router execution." Proceedings of the ACM SIGCOMM 2022.
- [FSE'26] Ning Kang, Peng Zhang, Jianyuan Zhang, Hao Li, **Dan Wang**, Zhenrong Gu, Weibo Lin, Shibiao Jiang, Zhu He, Xu Du, Longfei Chen, Jun Li, and Xiaohong Guan. "AccessRefinery: Fast Mining Concise Access Control Intents on Public Cloud." Proceedings of ACM FSE 2026.

Internship

- [2023 - 2025] ByteDance Inc, SDN.
 - Building a general distributed network verification framework on top of simulation-based network verifiers (*e.g.*, Batfish).

Ongoing Projects

- Versatile configuration migration planning via symbolic simulation and automated planning (AI Planning).
- Transient and convergence state reasoning of distributed network control plane through symbolic simulation.

Education

- [09/2017 - 06/2021] Computer Science and Technology, Xi'an Jiaotong University (XJTU), China
 - B.E.
- [09/2021 - 09/2026] Computer Science and Technology, Xi'an Jiaotong University (XJTU), China
 - Ph.D. student in NetVerify lab, **Advanced NeTwork System Group (ANTS)**. Advisor: Peng Zhang
- [07/2025 - 07/2026] Paul G. Allen School of Computer Science and Engineering, University of Washington, U.S.
 - Student intern in Computer Systems Lab. Advisor: Ratul Mahajan

Services

- ACM CoNEXT '26, Artifact Evaluation Committee.
- ACM SIGCOMM '26 Workshop on Formal Methods Aided Network Operation (FMANO '26), Program Committee.

- ACM/IEEE PACT'26, Artifact Evaluation Committee.
- ACM SIGCOMM '26, Artifact Evaluation Committee.